

Yelp Suspect Reviewer Detection

Unsupervised Low-Credibility Reviewer Detection on the Yelp Open Dataset

1,987,841

Total Users

3,439

Flagged (0.17%)

2

Independent Branches

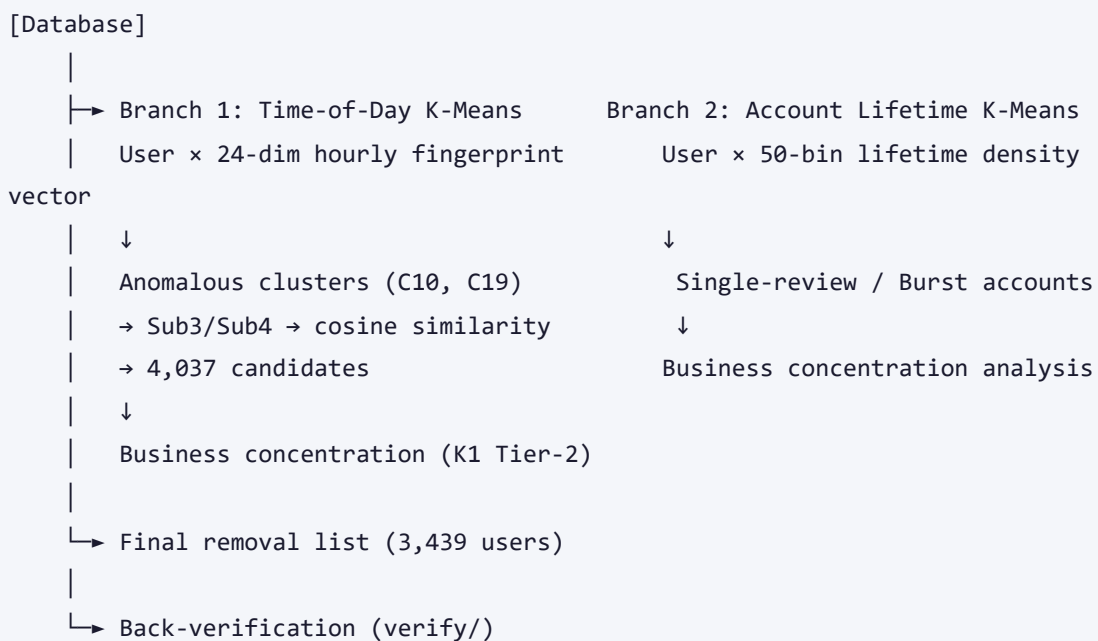
Unsupervised anomaly detection · Yelp Open Dataset · Side project

Yelp Suspicious User Detection

Overview

Unsupervised anomaly detection on the Yelp Open Dataset (1,987,841 users). No ground truth labels required — anomaly signals emerge from behavioral patterns across two independent dimensions, confirmed by business-level temporal clustering.

Pipeline



Branch 1 — Time-of-Day K-Means

Design

Human reviewers follow circadian rhythms. Accounts systematically posting at hours inconsistent with normal waking activity are behaviorally anomalous regardless of review

content.

Timestamp format: Yelp stores local time (empirically confirmed — PA users peak at 18:00 Eastern local time). Morning anomaly signal is timezone-accurate.

Features

Each user → **24-dimensional probability vector** of hourly review activity

- Per-year row normalization → summed → re-normalized (collapses year dimension)
- Users with < 5 total reviews excluded → **287,000 valid users**

Clustering & Anomalous Cluster Identification

MiniBatchKMeans (K=20), evaluated by **entropy** and **morning ratio (06:00–11:59)**.

Clusters 10 and 19 flagged: sharply elevated morning activity, near-zero during normal waking hours.

Cross-validation with business operating hours:

Sub-cluster	n	Morning reviews during open hours	Normal baseline
C10-Sub3	420	9%	43%
C10-Sub4	357	14%	43%

Cosine Similarity Scoring

Centroid built from Sub3+Sub4. Threshold = 5th percentile of known anomaly scores (**0.546**).
→ **4,037 anomaly candidates** (1.4% of 287k)

K1 Tier-2 Validation

Only **9 users** appear in both K1 candidates and the Lifetime removal list — the two branches are nearly orthogonal by design.

The remaining **4,028 K1-only users** (Tier 2) were validated using the same business concentration logic:

- 45 businesses showed coordinated bursts (span < 7d) → **78 users confirmed**, added to removal list
- No business reached suspicion_score ≥ 0.5

- closed_pct mean = 0.847 (vs. 0.721 for normal users) confirms morning anomaly is real, not a timezone artifact

Branch 2 — Account Lifetime K-Means

Design

Regardless of timezone, suspicious accounts cluster review activity at a specific point in their lifetime:

- **Single-review:** one review shortly after account creation, then silent
- **Burst-then-die:** dense early activity, then gone

Global Normalization

```
position = (review_date - yelping_since) / global_max_elapsed (~6,241 days)
```

Maps all users to a common [0, 1] timeline regardless of join date.

Single-Review Users (n = 1,135,945)

K-Means on 50-bin one-hot vectors.

Key finding: Cluster 1 (n=320,060), mean position ≈ 0.003

→ 28% of single-review users posted their only review within ~19 days of account creation.

Multi-Review Users (n = 851,896)

50-bin probability vectors, MiniBatchKMeans (K=20), evaluated by entropy.

Cluster	Entropy	Peak position	Pattern
1 (n=91,597)	0.31	0.00	Clearest burst-then-die

Burst definition: `peak_position  $\leq$  0.10` AND `entropy < 2.0`

→ Clusters 1, 5, 7, 9, 14 — **217,641 users**

Business Concentration & Temporal Clustering

Logic

High suspect coverage + short time window = coordinated campaign

High suspect coverage spread over years = natural accumulation (not a signal)

Metrics

Metric	Definition
<code>coverage_rate</code>	# suspect users / total reviews for that business
<code>span_days</code>	Days between first and last suspect review
<code>dense_30d_pct</code>	Fraction of suspect reviews in densest 30-day window
<code>suspicion_score</code>	$\text{coverage_rate} \times \text{dense_30d_pct}$

Coordination threshold: `span_days < 7`

High-overlap threshold: `suspicion_score ≥ 0.5`

Final Output

Dataset: **1,987,841 users**

File: `lifetime kmeans/result/final/removal_final.parquet`

Category	Count	% of Total	Criteria
Coordinated — single-review	1,424	0.072%	≥ 2 suspect accounts, same business, span $< 7d$
Coordinated — burst-cluster	744	0.037%	Same
Single-review, high overlap	1,193	0.060%	suspicion_score ≥ 0.5
K1 coordinated	78	0.004%	K1 candidate + business coordination
Total	3,439	0.173%	

Back-Verification: Methodological Independence

`verify/scripts/verify_all.py` checks the 3,439 flagged users across all analysis dimensions.

Metric	Value
Flagged users in K1 space (≥ 5 reviews)	514 / 3,439 (15%)
Avg cosine similarity of those 514	0.168 (K1 threshold = 0.546)
In Lifetime single-review system	2,617 (76%)
closed_pct — normal sample mean	0.721
closed_pct — flagged mean	0.740 ($\Delta = +0.018$)

Key insight: Flagged users look normal on dimensions they were *not* selected for. This isn't a problem — it is evidence that **each detection path found a genuinely distinct class of anomaly**, not the same population through different lenses:

Category	Behavioral Pattern	Inferred Motivation
single_coordinated	Multiple accounts, same business, 7-day window, then gone	Paid one-shot campaigns
burst_coordinated	Early burst, then silent; shares business targets	Bulk operations or acquired accounts
single_high_overlap	Reviews concentrated at suspect-heavy businesses	Ecosystem participants
k1_coordinated	Systematic morning posting + shared targets	Automated scripts or remote-operated accounts

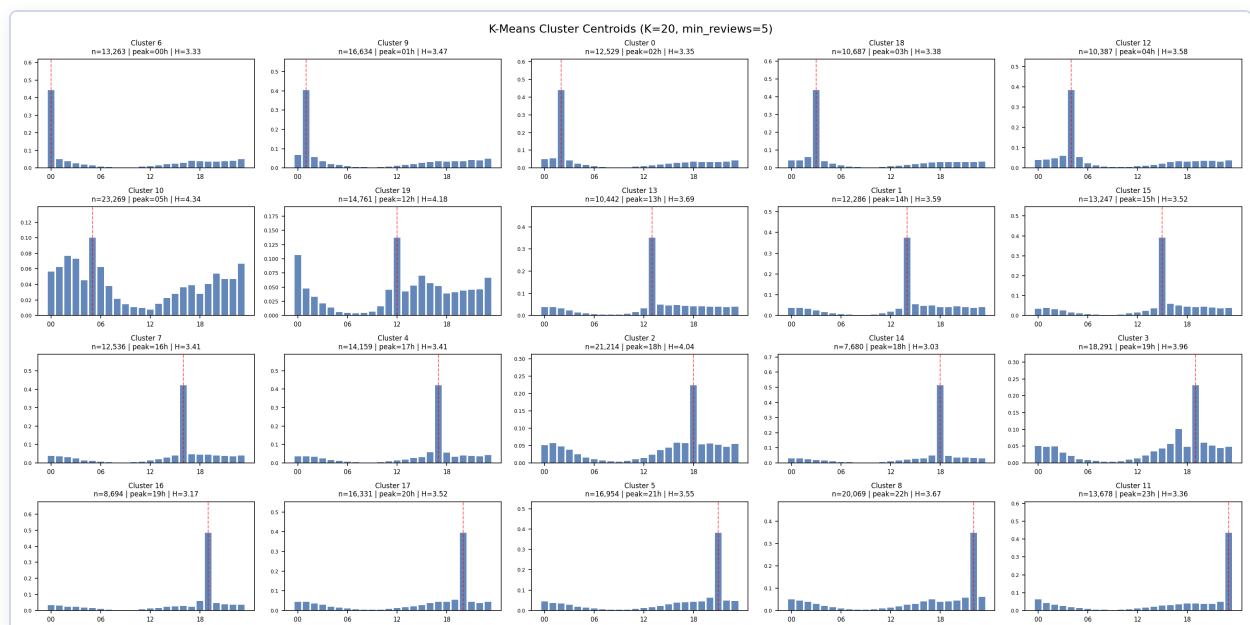
Threshold Rationale

Threshold	Reasoning
MIN_SUSPECT_USERS = 2	Two independent one-shot accounts hitting the same small business within 7 days is already implausible. Requiring 3 misses real coordination.
span_days < 7	3 days = near-certain coordination. 7 days still implausible as coincidence for new one-shot accounts. Beyond 14 days, organic word-of-mouth becomes a plausible explanation.
suspicion_score ≥ 0.5	Requires both ≥50% coverage AND ≥50% temporal concentration. Either condition alone is insufficient.
Burst users excluded without coordination	217,641 burst users share the same signature as early-adopter power users. Without direct coordination evidence, removal would be speculative. Only 744 linked to a 7-day burst are included.

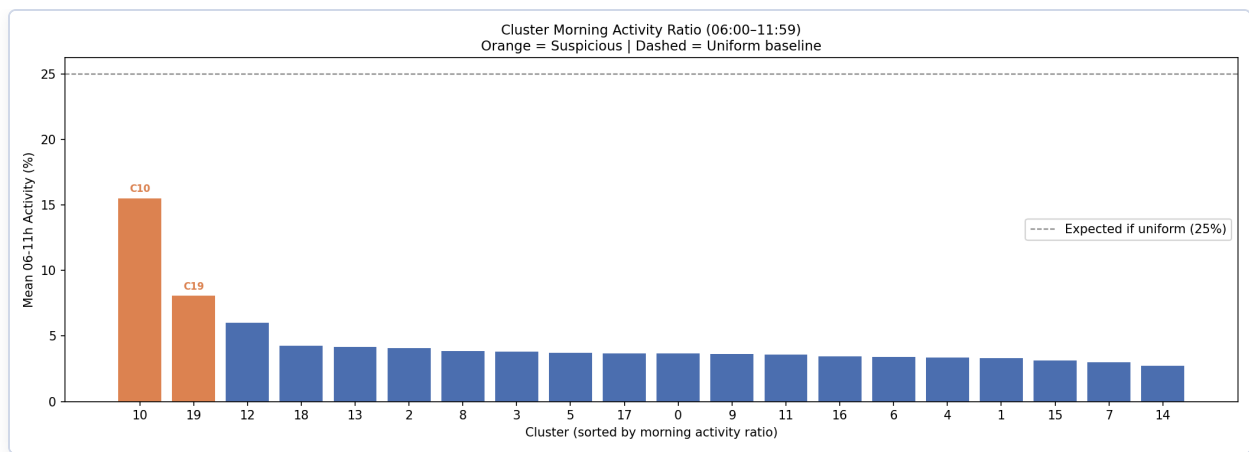
Detection Signal Summary

Signal	Branch	Strength
Morning activity 06–11h (local time)	Time-of-Day K-Means	Strong; cross-validated with business open hours
Business open-hours mismatch	Time-of-Day K-Means	Timezone-agnostic corroboration
Cosine similarity to anomaly centroid	Time-of-Day K-Means	Extends to all 287k users
K1 + business coordination	K-Means × Business concentration	Behavioral confirmation of K1 candidates
Single-review account	Lifetime K-Means	Broad; low precision alone
Burst-then-die lifetime	Lifetime K-Means	Moderate; strong when combined with clustering
Coordinated targeting (span < 7d)	Business concentration	Highest precision; smallest count

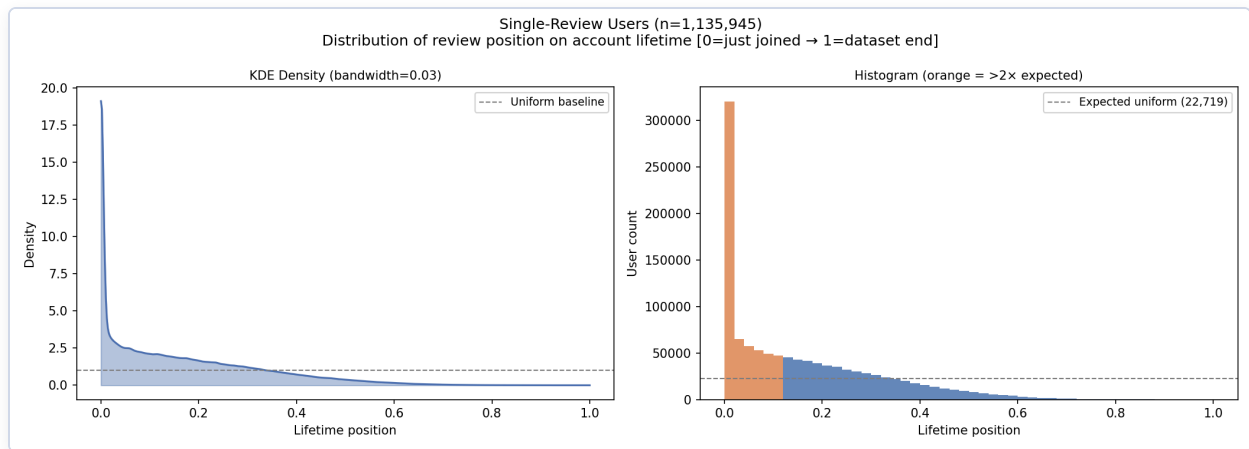
Key Visualizations



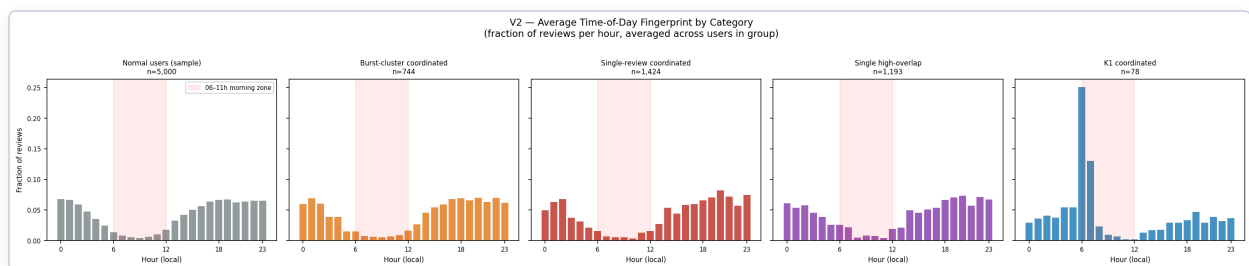
Branch 1 — Cluster Overview



Morning Review Ratio by Cluster



Branch 2 — Lifetime Position



Back-Verification: Hourly Fingerprint